

Safety of biologic agents for the management of rheumatic diseases during pregnancy.

PMID: 38456470 | Indexed in PubMed

To discuss the current understanding regarding the use of biologic therapeutics in pregnancy. Our understanding of the mechanisms underlying the potential fetal and infant exposure to biologics as well as a growing body of empirical evidence from real world use of biologics in pregnancy have demonstrated that biologics are generally compatible preconception and during pregnancy. Long-term effects of exposure to biologic agents in utero are not known, but will be uncovered in time. Biosimilars, which are becoming more popular, may not always share the same safety profiles as their originators. Biologics have revolutionized the management of rheumatologic disease and ushered in a new era of clinical remission among patients. These agents, developed and introduced into clinical use at the beginning of the new millennium, are very potent, yet their efficacy in treating disease often in reproductive aged women, raises questions regarding their safety during pregnancy. These therapeutics can cause immunosuppression and can inhibit immunologic circuits that are not only involved in disease pathophysiology but hypothetically could impact the development of the fetal immune system. Reassuringly, biologics, typically antibodies or antibody-based proteins, are introduced to the fetus via the typical route of transplacental antibody transfer, and thus only begin to be transferred in appreciable amounts in the second trimester (after organogenesis). From theoretic and empirical standpoints, biologic use during pregnancy appears well tolerated for fetal development and to not substantially affect infant immune development.

Safety of Biologics for Atopic Diseases During Pregnancy.

PMID: 35987486 | Indexed in PubMed

The high prevalence of atopic diseases in women of childbearing age reveals the need to determine the safety of biologics during pregnancy. This review summarizes the effects of 7 Food and Drug Administration-approved biologics (omalizumab, mepolizumab, reslizumab, benralizumab, dupilumab, tezepelumab, and tralokinumab) on maternal and fetal outcomes. For this purpose, we reviewed English-language publications to investigate whether the use of biologics for atopic diseases during pregnancy increased the risk of preterm delivery, stillbirth, low birth weight, or congenital malformations. Most publications found were case reports, case series, or observational studies reporting outcomes in a total of 313 pregnancies. No randomized controlled studies were identified. We found that biologics do not seem to influence maternal or fetal outcomes. Indeed, worsening of the underlying atopic disease during pregnancy appears to be more detrimental to the viability of the pregnancy. Given the small sample size and scarcity of studies, future research should include prospective studies with comparable control groups without exposure to biologics and multicenter registries for long-term follow-up.

[Biologics during pregnancy and breast-feeding].

PMID: 21063826 | Indexed in PubMed

Over the past years, biological therapies, especially anti-TNF- α antibody therapy has emerged as a treatment approach in patients who have failed to achieve or maintain remission with traditional DMARDs. Women suffering from inflammatory arthritis may need to continue therapy throughout pregnancy and/or

in the lactation period, hence the increased concern over the safety of antirheumatic drugs during pregnancy. Anti-TNF agents fall within the US FDA category B concerning fetal risk, indicating that no adequate and well-controlled studies have been conducted in pregnant or lactating women. However, in the last decade, numerous case series and registry data of pregnancies exposed to anti-TNF therapy have accumulated in the literature. According to these data, TNF inhibitor therapies appear to be safe in pregnancy, since no increased risk of malformations has been demonstrated. Ceasing therapy after conception should be considered, but treatment may be continued during pregnancy when indicated. The use of these agents is likely compatible with breast-feeding. The extent of fetal risk is not clarified for exposure to other biologics, such as Rituximab.

The use of biologics and small molecules in pregnant patients with rheumatic diseases.

PMID: 30227748 | Indexed in PubMed

Biological agents have radically changed the prognosis of rheumatic patients. Current evidence demonstrates that tight disease control during pregnancy is mandatory to minimize adverse outcome risk. As the new therapeutic tools are pivotal to maintain appropriate disease activity, it is timely to review available evidence about the safety of biologics and small molecules in pregnancy. Areas covered: A comprehensive literature review has been performed, reporting available data about the passage into breast milk, rate of pregnancy loss and fetal malformations, and long-term complications due to in utero exposure to biological agents and small molecules. Expert commentary: Data about the safety of agents against tumor necrosis factor in pregnancy are reassuring. Even rituximab, tocilizumab, belimumab, ustekinumab, secukinumab, and abatacept have not been associated with an increased rate of fetal abnormalities or adverse pregnancy outcome. Experience with small molecules is too small to draw any conclusion. Even if further data are warranted to define the possible long-term effects of in utero biologic exposure on the infant immune system development, it is reasonable to speculate that in the next future the use of biologics during pregnancy will continue to expand, at least when maternal benefit justifies the potential risk to the fetus.

Le choix thérapeutique : médicaments du psoriasis et grossesse avant, pendant et après la grossesse.

PMID: 33242008 | Indexed in PubMed

Psoriasis typically affects young adults and therefore many women with a desire to become pregnant or already pregnant. In this particular situation, treatment can be a real challenge for some patients, especially in the case of severe forms. In addition to local treatments, which are generally well tolerated, UVB phototherapy and cyclosporin remain the first-line systemic treatments. Acitretin and methotrexate are contraindicated. Safety data regarding the administration of biologic agents during pregnancy, are reassuring, the main adverse event being immunosuppression of the newborn if treatment is not discontinued. Biologic agents should ideally be discontinued before pregnancy, but in case of absolute necessity, they can be maintained or even initiated during pregnancy. Overall, it is recommended that biologic agents should not be continued beyond the second trimester because of the risk of maternal-fetal infection. If a biologic agent should be initiated during pregnancy, tanercept or certolizumab will be preferred, because of their low transplacental passage and more extensive safety

data.

Prospectively-followed pregnancies in patients with inflammatory arthritis taking biological drugs: an Italian multicentre study.

PMID: 26311348 | Indexed in PubMed

Information on new drugs does not include their possible effects on pregnancy because pregnant women are excluded from clinical trials. Although not classified as teratogenic in animals, limited data is available on biological anti-rheumatic agents and their safety in human pregnancy. The aim of the study is to evaluate the safety of biological drugs in pregnant patients with chronic arthritis. Pregnancy outcome and maternal disease variations were prospectively followed in six Italian Rheumatology Centres. Patients exposed to biological agents during the periconceptional period or during pregnancy were included in the study. The occurrence of congenital malformations as well as the obstetric and neonatal outcomes were assessed. Between 1999 and 2013 we identified 79 exposed pregnancies in 67 women affected by different rheumatic diseases with peripheral chronic arthritis. At the time of the start of pregnancy, 56 patients were taking etanercept, 13 adalimumab, 3 infliximab, 2 each certolizumab-pegol and rituximab, 1 each golimumab, anakinra and abatacept. Biological treatment was stopped after a mean of 41 days since documented pregnancy. Live births were reported in 66% of pregnancies. The rate of spontaneous pregnancy loss was 20%. Only one congenital malformation was reported. TNF-alpha inhibitors can be considered safe in the periconception period, representing a possible therapeutic choice also in young women affected by an aggressive form of chronic arthritis and hoping for a pregnancy. Reports of exposure during 2nd/3rd trimester are still limited and suggest caution. Experience with abatacept, tocilizumab, anakinra and rituximab in pregnancy is insufficient.

Biologic therapies and pregnancy: the story so far.

PMID: 24352337 | Indexed in PubMed

Biologic therapies have revolutionized treatment outcomes for patients with inflammatory arthritis. However, there remains a concern regarding their safety during conception, pregnancy and breastfeeding. Data on the safety of these treatments are largely limited to uncontrolled case reports. Collective evidence from many hundreds of pregnancies in inflammatory arthritis and IBD have suggested that exposure to anti-TNF therapies at the time of conception or during the first trimester does not result in an increased risk of adverse pregnancy and fetal outcomes. Monoclonal antibodies, and to a lesser extent recombinant fusion proteins, do cross the placenta during the second and third trimester and are functional in the fetus, as evidence by lymphopaenia reported at birth in children exposed to rituximab in utero. In addition, live vaccines should be avoided in children with in utero exposure to biologics for at least the first 6 months of life. The longer-term effects of in utero exposure remain unknown. Studies suggest that many of these drugs do enter breast milk in small amounts, but the extent to which they are absorbed by the infant is less clear. Limited reports have not suggested adverse pregnancy outcomes in women whose partners were exposed to anti-TNF therapies or rituximab at the time of conception. Data on other biologic therapies, including anakinra, abatacept and tocilizumab, in both men and women remain extremely limited.

The London Position Statement of the World Congress of Gastroenterology on Biological Therapy for IBD with the European Crohn's and Colitis Organisation: pregnancy and pediatrics.

PMID: 21157441 | Indexed in PubMed

Women with inflammatory bowel disease (IBD) have similar rates of fertility to the general population, but have an increased rate of adverse pregnancy outcomes compared with the general population, which may be worsened by disease activity. Infertility is increased in those undergoing ileal pouch-anal anastomosis. Anti-tumor necrosis factor therapy in pregnancy is considered to be low risk and compatible with use during conception in men and women and during pregnancy in at least the first two trimesters. Infliximab (IFX) and certolizumab pegol are also compatible with breastfeeding, but safety data for adalimumab (ADA) are awaited. The safety of natalizumab during pregnancy is unknown. For children with Crohn's disease (CD), IFX is effective at inducing and maintaining remission. Episodic therapy is not as effective as scheduled infusions. Disease duration in children does not appear to affect the efficacy of IFX. IFX promotes growth in prepubertal and early pubertal Crohn's patients. It is also effective for the treatment of extraintestinal manifestations. ADA is effective for children with active CD and for maintaining remission, even if they have lost response to IFX, although there are fewer data. Vaccination of infants exposed to biological therapy in utero should be given at standard schedules during the first 6 months of life, except for live-virus vaccines such as rotavirus. Inactivated vaccines may be safely administered to children with IBD, even when immunocompromised.

Management of the pregnant inflammatory bowel disease patient on anti-tumour necrosis factor: state of the art and future directions.

PMID: 25101334 | Indexed in PubMed

Antitumour necrosis factor (anti-TNF) therapy has been a major advance in the treatment of inflammatory bowel disease (IBD) by improving rates of mucosal healing, steroid-free remission, and decreasing rates of hospitalization and surgery. Because IBD affects women in their reproductive years, clinicians have and will continue to be asked in the future about the safety profile of these agents and their potential impact on pregnancy, the developing fetus and newborn. Immunoglobulin G transfer from the mother to fetus begins in the second trimester, with an elevation starting at 22 weeks of gestation and the largest amount transferred in the third trimester. Although research investigating the long-term outcomes of children exposed to anti-TNF therapy in utero is limited, there is no known adverse effect on either pregnancy or newborn outcomes including infectious complications with this class of drugs. The World Congress of Gastroenterology consensus statement on biological therapy for IBD considered infliximab and adalimumab to be low risk and compatible with use during conception and during pregnancy in at least the first two trimesters. Based on a clinical algorithm used at the University of Calgary Pregnancy and IBD clinic (Calgary, Alberta), recommendations have been provided on the management of pregnant patients on anti-TNF therapy, particularly with regard to third-trimester dosing, taking into account disease characteristics of individual patients. When educated about the safety of anti-TNF therapy during pregnancy, patients often choose to continue on therapy during the third trimester.

Psoriasis in pregnancy: a review (II).

PMID: 24314892 | Indexed in PubMed

Scarce scientific evidence is available to define the precise effects that certain drugs might have on embryonic and fetal development if taken by pregnant women with psoriasis, given the ethical concerns that preclude enrolling such women in clinical trials. The little information on the use of biologics during gestation that has been published is based on retrospective and observational studies, and experience with these drugs in this context in psoriasis is still very limited. The literature seems to suggest that biologic therapy is safe during pregnancy, but there is no certainty. This detailed review of accumulated experience with biologic therapy during pregnancy relies mainly on descriptions of the management of other types of rheumatic disease, although the use of these agents in psoriasis is growing steadily.
